Assignment 4: Fingerprint Quality Assessment Report

Q1. What threshold did you set for blur? How did you decide?

Answer: The Laplacian variance blur threshold was set to **8.0** based on empirical calibration across the dataset. Clear images consistently produced scores between **8.29** and **11.26**, while motion-blurred images consistently dropped below **8.0** (reaching as low as **2.78**).

Q2. Which metric was hardest to implement correctly? What went wrong first?

Answer: Glare Detection and **Ridge Clarity** within the region of interest were the most challenging. Initially, running glare calculations across the whole unmasked frame evaluated normal background highlights as overexposed pixels, leading to false rejections. Restricting calculations exclusively inside the segmented **Finger ROI Mask** resolved the issue.

Q3. What is NFIQ2? Why is a score designed for contact scanners not reliable for phone camera images?

Answer: NFIQ2 (NIST Fingerprint Image Quality 2) is a standard quality scoring tool trained on 500 DPI contact optical/capacitive scanner images. It is unreliable for contactless phone camera images because phone captures introduce variable distance, focal blur, non-uniform ambient lighting, and perspective tilt, causing NFIQ2 to penalize otherwise usable contactless captures.

Q4. Name 3 other quality problems you'd add checks for in a real deployment.

Answer:
1. **Distance & Scale Check:** Verifying finger pixel width to ensure the finger is placed at a valid distance.
2. **Perspective & Pitch Angle:** Detecting out-of-plane rotation to keep the finger pad parallel to the camera.
3. **Lens Contamination Detection:** Identifying smudges, moisture, or grease obscuring ridge detail.

Q5. If a rural agricultural worker's fingerprints are naturally worn, what should the system do differently?

Answer:
1. **Adaptive Thresholds:** Dynamically relax ridge clarity thresholds after failed attempts while applying targeted contrast enhancement (CLAHE).
2. **Multi-Frame Burst Stacking:** Capture a rapid sequence of frames to average noise and amplify faint ridge structures.
3. **Alternative Modality Fallback:** Automatically offer fallback verification via secondary fingers or facial recognition to prevent biometric exclusion.